Ethereum 30-Day Network Activity Analysis

Analysis window: August 24 to September 23, 2026 (UTC)
Daily comparisons: August 25 to September 22, 2026
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Executive summary

Ethereum sustained a broadly consistent level of transaction activity during the observation window. Across the 29 complete UTC days, the network recorded 55.15 million transactions, averaging 1.90 million per day. Daily transaction counts moved closely with active-address counts, while gas consumption and block gas utilization varied very little. Gas prices fluctuated considerably more than computational usage.

Complete-day metric	Result
Transactions recorded	55,151,035
Mean daily transactions	1,901,760
Mean daily active addresses	635,690
Mean transactions per active address	3.014
Mean block gas utilization	50.61%
Transaction-weighted execution success	98.49%
Transactions with non-removed logs	63.51%

Key findings

- Transaction activity ended 4.98% above the first complete day, while active addresses ended 8.67% higher. These endpoint changes do not establish a sustained growth trend.
- Gas consumption was stable: its daily coefficient of variation was 0.24%, compared with 17.18% for daily mean gas price.
- September 13 was the broadest anomaly under the five-metric screen: transaction count, active addresses, and transactions per block were unusually low.
- Receipt-level execution succeeded in 54,319,998 of 55,151,035 transactions. Logs provided observable contract-event evidence for 35,024,739 transactions.

The evidence describes transaction throughput, address participation, computational usage, and execution outcomes. It does not establish unique-user growth, economic value transferred, or the causes of individual daily movements.

Scope, data, and methodology

The analytical question is: What happened on Ethereum during this 30-day period, and what does the data reveal about network activity? The workflow separates blockchain-scale aggregation in GoogleSQL from daily-level analysis in Python/Pandas.

Window metadata	Recorded value
Analysis start (inclusive)	2026-08-24 10:01:35 UTC
Analysis end (inclusive)	2026-09-23 10:01:35 UTC
Recorded starting block	25,824,304
Recorded latest indexed / ending block	26,039,452
Recorded latest indexed block timestamp	2026-09-23 10:01:35 UTC
Elapsed window / calendar dates	30 days / 31 dates
Comparable complete-day sample	29 days: August 25 through September 22

Data source and workflow

Primary dataset: bigquery-public-data.goog_blockchain_ethereum_mainnet_us. Transactions and blocks provide activity and gas measures; receipts provide execution status; logs and traces provide observable contract-activity measures. The block metadata is recorded in the project rather than presented as the current chain tip.

Ethereum Mainnet → BigQuery / GoogleSQL → daily CSV exports → Python / Pandas → statistics and visualizations → interpretation. SQL filters the timestamp window and aggregates before export. Transaction intensity, anomaly detection, and correlations reuse existing outputs, avoiding additional raw-table scans.

Validation and statistical conventions

Each of the six exported datasets contains 31 unique, consecutive dates, with no null values. Transaction counts match receipt and contract-activity totals on every date. Successful plus failed executions reconcile to transaction totals. Gas measures agree between the gas and utilization exports within floating-point precision. These checks validate the supplied analytical outputs; they do not substitute for a fresh source query or transaction-level audit.

August 24 and September 23 are partial days and remain in full-window totals. All daily means, medians, ranges, rolling averages, anomaly baselines, and correlations in this report use the 29 complete days. Standard deviation uses the sample convention (ddof = 1). Period success and contract-activity shares use summed counts; daily mean rates give each day equal weight.

Tools used: Google BigQuery, GoogleSQL, Python, Pandas, Matplotlib, Plotly, Jupyter notebooks, and Git.

Ethereum Transaction Activity and Active Addresses

Transaction throughput fluctuated around a stable daily baseline. Across the exact timestamp window, including both partial days, there were 56,994,725 transactions. The 29 complete days accounted for 55,151,035 transactions.

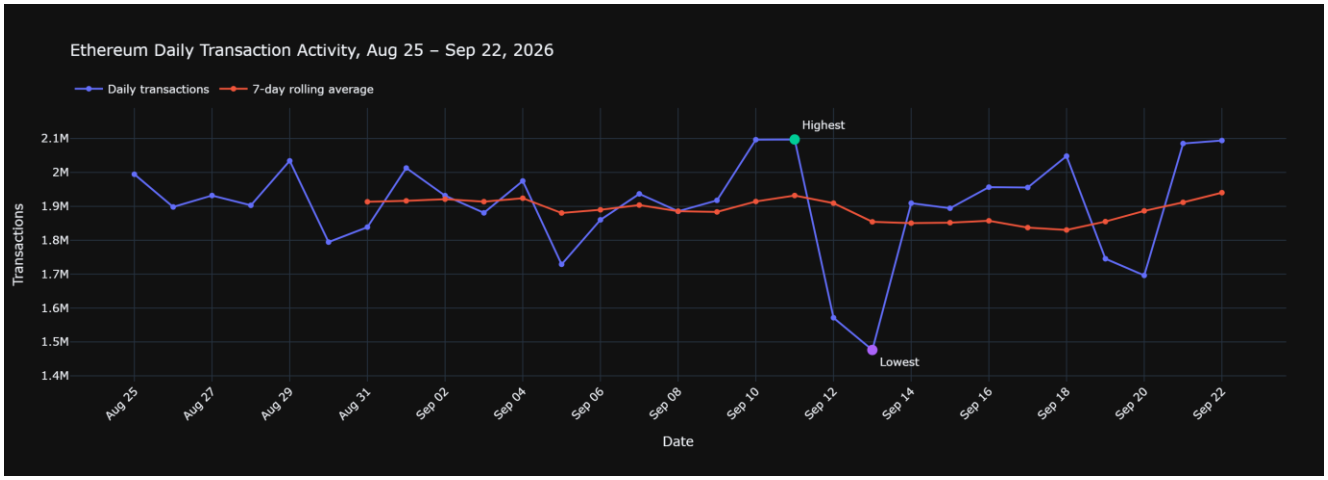


Figure 1. Daily transaction counts and trailing 7-day mean. Complete UTC days only.

Metric	Transaction count	Active addresses
Daily mean	1,901,760	635,690
Daily median	1,917,517	645,948
Highest day	Sep 11: 2,097,342	Sep 10: 829,303
Lowest day	Sep 13: 1,476,230	Sep 13: 454,329
First-to-last change	+4.98%	+8.67%
Coefficient of variation	7.87%	12.86%

The first and last complete-day transaction counts were 1,994,723 and 2,094,087. The highest daily count was 42.07% above the lowest. A higher ending value alongside a 7.87% coefficient of variation supports a description of fluctuating throughput, rather than a continuous upward trajectory.

Address participation

An active address is a distinct, non-null address appearing as a transaction sender or recipient on a given day; each address is counted once that day. Active-address counts varied more than transaction counts and rose from 672,803 on August 25 to 731,146 on September 22. The September 13 low occurred in both series.

Active addresses include contract addresses and are not unique users. A daily union of senders and recipients also differs from a sender-only measure. Daily distinct counts cannot be summed to obtain distinct addresses across the entire period.

Transaction Intensity

Transaction intensity is daily transaction count divided by daily active addresses. It connects transaction throughput with the breadth of address participation, using the same sender-or-recipient definition as Task 2.

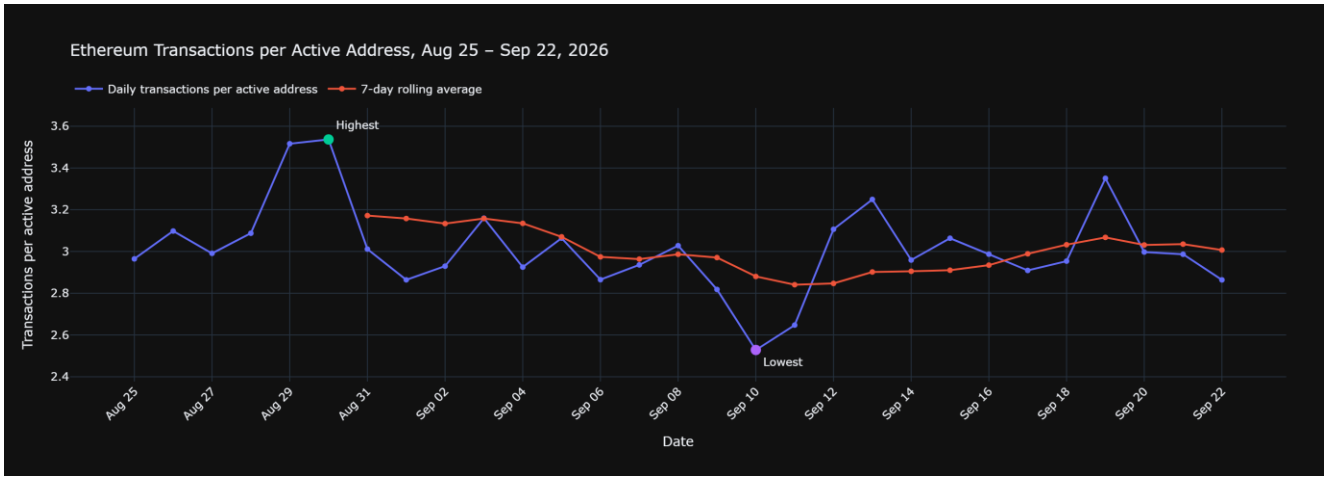


Figure 2. Daily transactions per active address and trailing 7-day mean.

Statistic	Transactions per active address
Mean / median of daily ratios	3.014 / 2.987
Highest complete day	August 30: 3.537
Lowest complete day	September 10: 2.528
First complete day	August 25: 2.965
Last complete day	September 22: 2.864
First-to-last change	-3.40%
Standard deviation / coefficient of variation	0.212 / 7.04%

Interpretation

The ratio remained close to three transactions per active address. Its decline between the two endpoints reflects active-address counts increasing more than transaction counts over those endpoints. The lowest ratio occurred on September 10, the day with the highest active-address count.

This is a network-level ratio, not a measured average number of transactions sent by each address. The denominator includes recipients and contracts, while each transaction is counted once in the numerator. It cannot establish individual user engagement, the distribution of address-level activity, or how many transactions a typical user initiated.

The reported mean is the arithmetic mean of 29 daily ratios. It is distinct from dividing period transaction totals by summed daily active-address counts, and neither produces a period-wide unique-user measure.

Network Utilization

Daily transactions per block varied considerably more than gas utilization. The utilization query computes the daily arithmetic mean of each block's gas_used / gas_limit ratio, expressed as a percentage.

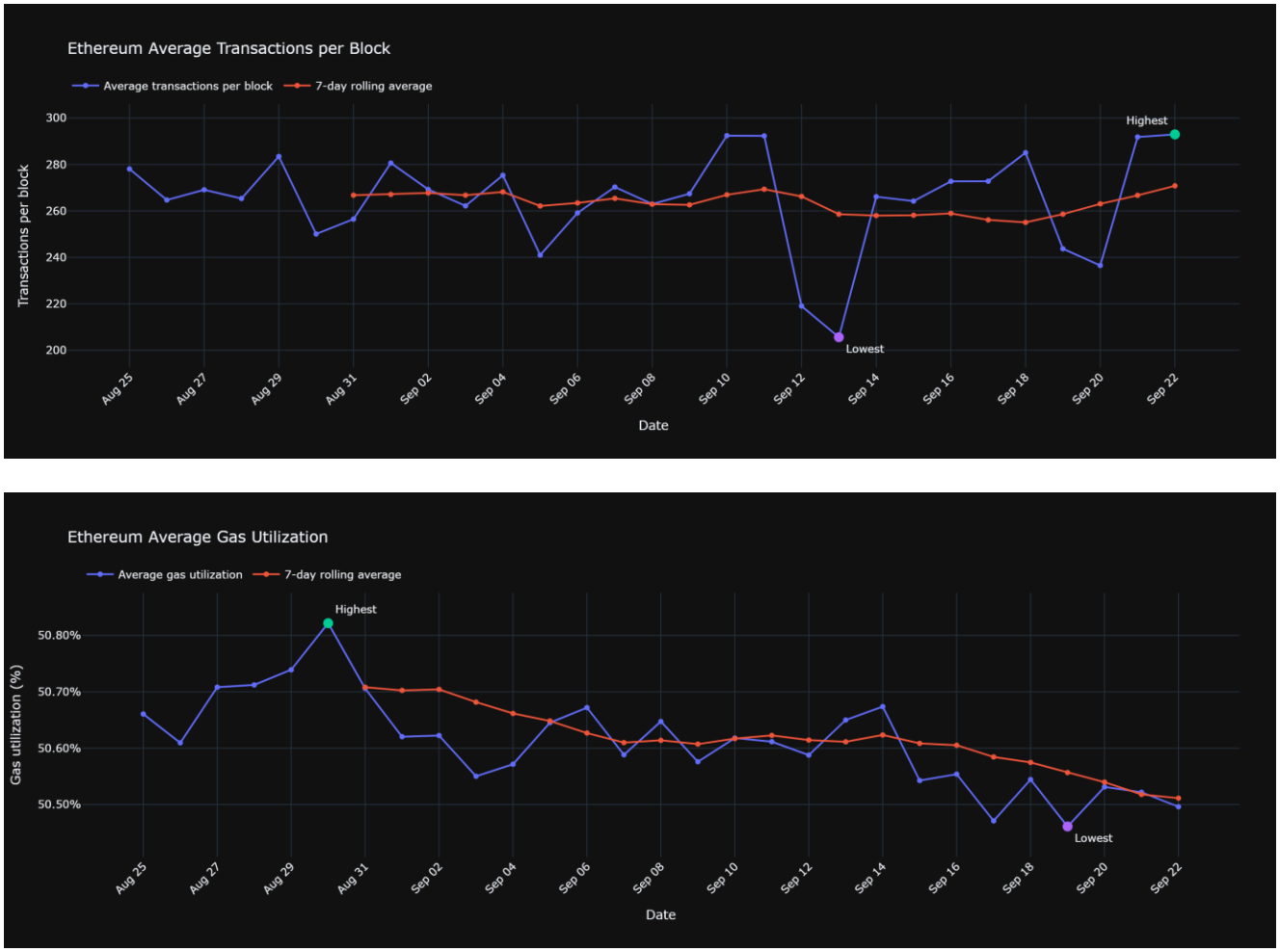


Figure 3. Transactions per block and gas utilization.

Metric	Mean	Complete-day range
Transactions per block	265.18	205.57–292.96
Daily gas consumed	217.77 billion gas	216.55–218.83 billion gas
Block gas utilization	50.61%	50.46%–50.82%
Mean daily block gas limit	approximately 60.00 million gas	approximately 60.00 million gas

Transactions per block peaked on September 22 and were lowest on September 13. Gas utilization peaked on August 30 and was lowest on September 19. Their coefficients of variation were 7.90% and 0.16%, respectively.

Interpretation

The network recorded different transaction counts while consuming a similar amount of gas each day. Transaction count therefore provides a different view of activity from computational usage: transactions do not all consume equal gas.

Utilization of about 50.61% describes gas consumed relative to block gas limits. It does not by itself establish congestion, unused economic capacity, transaction waiting times, or fee affordability. Daily averages can conceal variation between individual blocks.

The daily utilization measure is a mean of block-level ratios. It should not be relabeled as total daily gas used divided by total daily gas limits, nor as a transaction-weighted utilization measure.

05 | TASK 5

Gas Market Activity

Gas prices fluctuated much more than gas consumption. Daily mean gas price averaged 0.815 Gwei and ranged from 0.465 Gwei on August 29 to 1.148 Gwei on September 21.

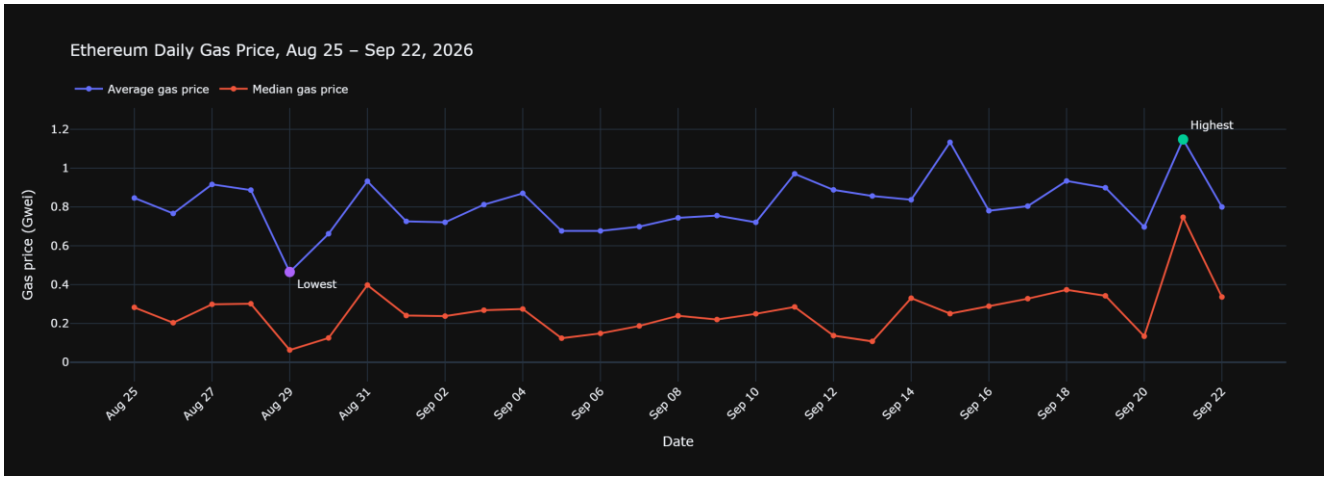


Figure 4. Daily mean and approximate median gas price.

Measure	Result
Mean of daily mean gas prices	0.815 Gwei
Median of daily mean gas prices	0.804 Gwei
Mean of approximate daily median prices	0.259 Gwei
Daily mean-price coefficient of variation	17.18%
Daily gas-consumption coefficient of variation	0.24%
Daily utilization coefficient of variation	0.16%

Relatively high-price days

Using the 90th percentile of daily mean gas price (0.941 Gwei) as a descriptive threshold identifies September 11 (0.971 Gwei), September 15 (1.133 Gwei), and September 21 (1.148 Gwei). This ranks prices within the sample; it is separate from the two-standard-deviation anomaly screen.

Daily mean prices exceed approximate daily medians throughout the complete-day sample. That is consistent with higher-price observations pulling the mean upward, although daily summaries do not reveal the full transaction-level price distribution.

Measurement limits

The query aggregates the transactions table's gas_price field and converts Wei to Gwei by dividing by 1 billion. It does not separately measure base fees, priority fees, effective paid fees by transaction type, ETH expenditure, or USD transaction costs. The median is estimated with APPROX_QUANTILES, not calculated as an exact transaction-level median. The mean of daily means is not a transaction-weighted period gas price.

06 | TASK 6

Transaction Success

Receipt status indicates predominantly successful execution. Across the 29 complete days, 54,319,998 transactions succeeded and 831,037 failed, reconciling to 55,151,035 transactions.

Execution outcome	Transactions	Period share
Successful (status = 1)	54,319,998	98.49%
Failed (status = 0)	831,037	1.51%
Total	55,151,035	100.00%

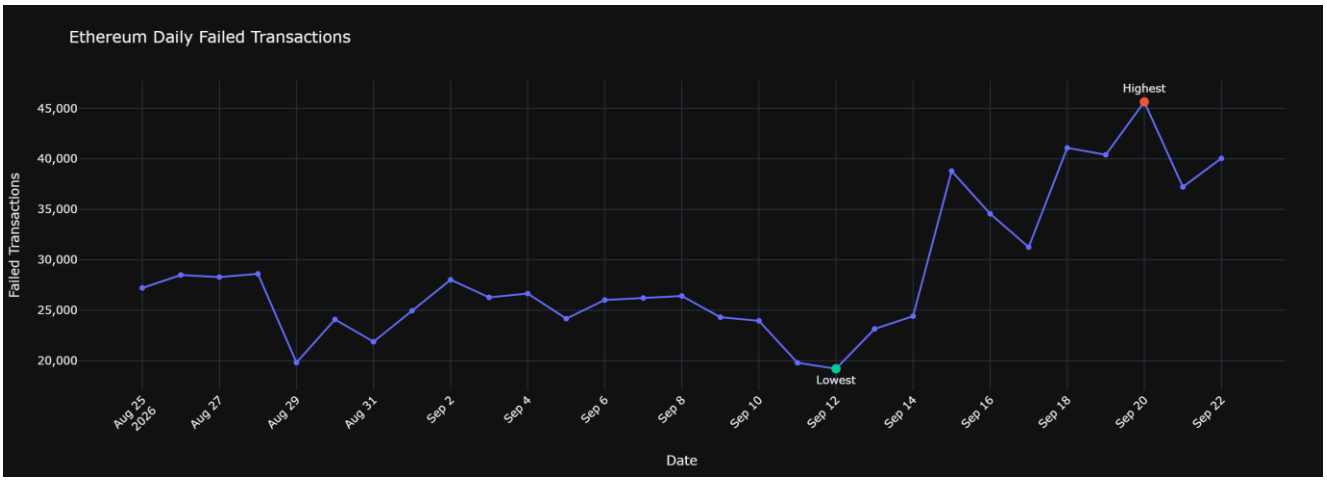


Figure 5. Daily receipt-level failure rate

Daily variation

The lowest failure rate was 0.94% on September 11; the highest was 2.69% on September 20. The arithmetic mean of daily failure rates was 1.513%, while the transaction-weighted period rate was 1.507%. The latter is the reported overall 1.51% failure rate.

The largest day-over-day rise was +0.769 percentage points on September 15. The largest fall was -0.906 percentage points on September 21. The notebook screens daily rate changes against their mean plus or minus two sample standard deviations; these changes identify investigation candidates rather than failure causes.

Interpretation

Execution success was the dominant outcome, including on higher-failure days. Receipt failure is not a measure of chain downtime, dropped transactions, or transactions never included in a block. These data do not distinguish specific causes such as reverts or gas exhaustion, and successful execution does not prove that a transaction achieved an intended economic outcome.

Contract vs EOA Activity

The analysis measures observable contract-related activity using logs and CREATE traces. It does not perform a complete historical recipient-level classification of every transaction as EOA-to-EOA or contract-directed.

Observable measure	Transactions	Period share
At least one non-removed log	35,024,739	63.51%
No observed non-removed log	20,126,296	36.49%
At least one CREATE trace	299,259	0.54%



Figure 6. Daily shares of transactions with logs and CREATE traces. These measures can overlap and are shown separately.

The daily log-observation rate averaged 63.53%, ranging from 47.79% on August 29 to 66.46% on September 8. The CREATE-trace rate averaged 0.54%, ranging from 0.45% on September 19 to 0.62% on September 8.

Classification boundaries

Logs are evidence of observable contract-event activity, but a contract interaction can occur without emitting logs. Transactions without observed logs therefore were not classified as EOA-to-EOA activity. The 63.51% measure is a share with observed logs, not an exhaustive share of all contract interactions.

CREATE traces can include internal creation activity, and transactions with CREATE traces may also emit logs. The two shares must not be added. The SQL does not filter CREATE traces by successful execution, so the 0.54% measure represents transactions associated with creation traces, not a verified count of successfully deployed contracts.

The evidence supports a substantial presence of observable contract activity. It does not establish the relative market shares of DeFi, NFTs, payments, or other application categories.

08 | TASK 8

Network Activity Anomalies

Five daily metrics are standardized against the 29-complete-day baseline: transaction count, active addresses, transactions per block, total gas used, and mean gas price. A metric is flagged when |z| ≥ 2, where z = (daily value - sample mean) / sample standard deviation.

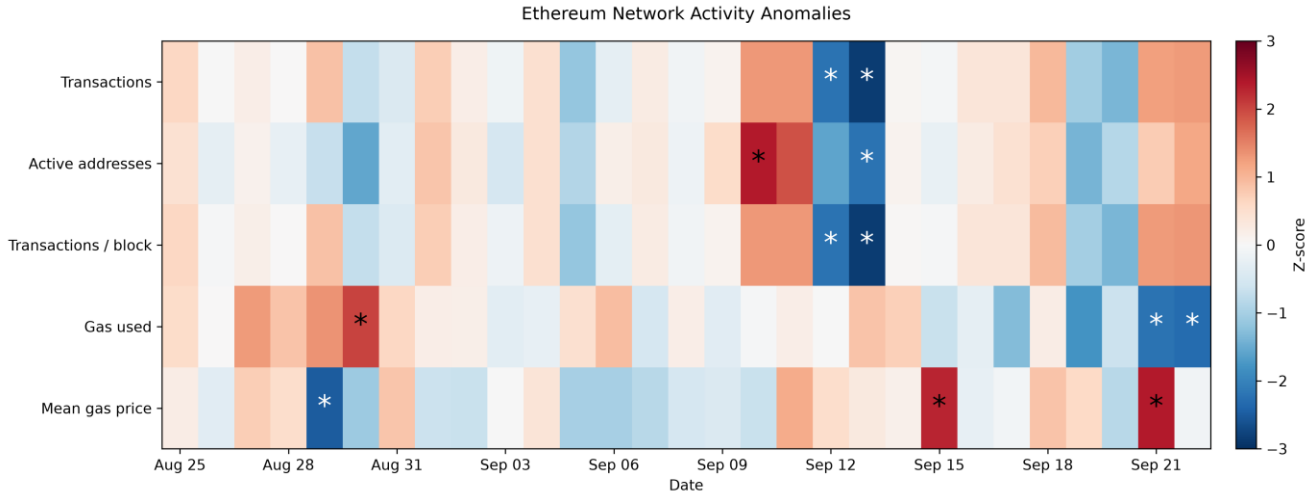


Figure 7. Five-metric z-score screen. Asterisks mark |z| ≥ 2; red indicates above-baseline values and blue indicates below-baseline values.

Figure 7. Daily z-score heatmap for five Ethereum network metrics. Asterisks mark |z| ≥ 2; red indicates values above each metric's 29-day mean, and blue indicates values below it.

Date	Flagged metrics and direction	z-scores
Aug 29	Mean gas price: low	-2.500
Aug 30	Gas used: high	+2.032
Sep 10	Active addresses: high	+2.368
Sep 12	Transactions / transactions per block: low	-2.210 / -2.205
Sep 13	Transactions / addresses / tx per block: low	-2.845 / -2.218 / -2.845
Sep 15	Mean gas price: high	+2.277
Sep 21	Gas used: low / mean gas price: high	-2.209 / +2.381
Sep 22	Gas used: low	-2.307

Eight days had at least one flagged metric. September 12, September 13, and September 21 had at least two. September 13 had the most flags, with three unusually low activity measures. September 21 combined a relatively low gas-consumption observation with a relatively high mean gas price.

How to interpret the flags

Transaction count and transactions per block have correlation r = 0.9999 in this sample. Their simultaneous flags are largely redundant, so multiple flags do not necessarily provide independent evidence of a broad network shock. A gas-usage flag can also reflect a small absolute change because that series is exceptionally stable.

This is an exploratory threshold screen, not a significance test or event-attribution method. The short sample, dependent metrics, and unmodeled calendar effects limit generalization. Unusual observations may be high or low; the screen does not measure only upward spikes.

09 | TASK 9

Relationships Between Metrics

Pearson correlations quantify linear association across 29 complete daily observations. Transaction count had a strong positive association with active addresses, but little linear association with daily mean gas price.

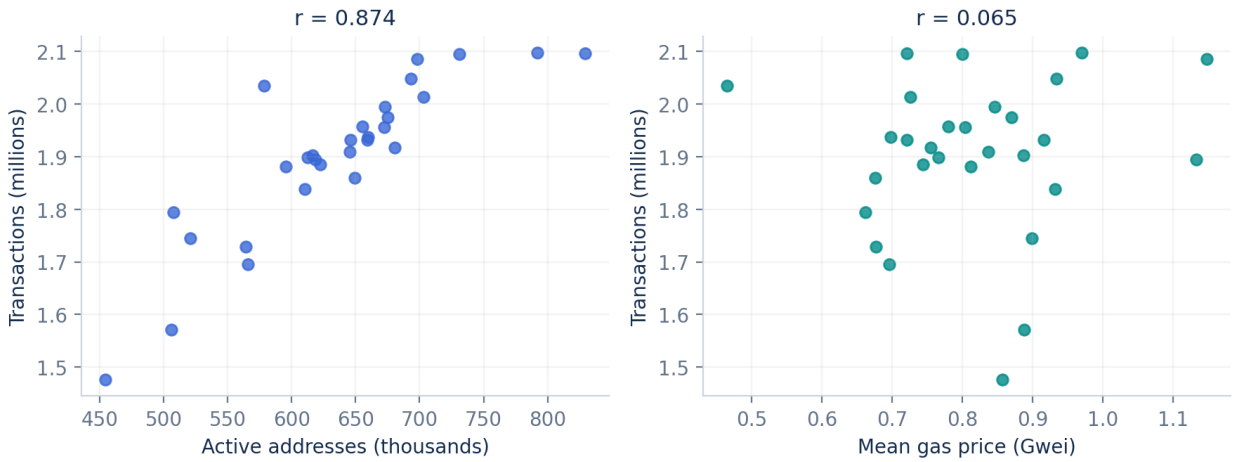


Figure 8. Daily scatter plots compare the strongest specified positive relationship with the weakest specified linear relationship.

Relationship	Pearson r	Interpretation
Transactions vs active addresses	+0.874	Strong positive
Active addresses vs transaction intensity	-0.839	Strong negative; shared denominator
Transactions vs total gas used	-0.223	Weak negative
Gas utilization vs transactions	-0.115	Weak negative
Mean gas price vs transactions	+0.065	Little linear association

Analytical meaning

Higher transaction-count days tended to have more active addresses. The weak relationships with gas consumption and gas utilization are consistent with transactions varying in computational cost. The near-zero gas-price relationship indicates little daily linear co-movement in this sample; it does not establish that gas prices are independent of network activity at all timescales.

The intensity metric contains active addresses in its denominator, creating mathematical dependence. Its -0.839 correlation with active addresses should not be presented as an independent behavioral discovery. Similarly, transaction count and transactions per block share an underlying count, which explains why they move almost identically.

Correlation does not establish causation. Daily aggregation can hide hourly relationships, and Pearson correlation can miss nonlinear patterns. With 29 observations and no reported inferential testing, the results should remain descriptive and specific to the analysis window.

Synthesis, limitations, and next steps

What happened during the window?

Ethereum processed approximately 57.00 million transactions in the exact 30-day window. Within the comparable 29-day sample, daily activity averaged 1.90 million transactions and 635,690 active addresses. Both series ended above their first complete-day values, but fluctuated enough that endpoint increases should not be treated as proof of sustained growth.

Address participation and transaction throughput moved together, while gas usage stayed nearly constant. The gas market varied more in price than in consumed computation. Receipt outcomes were overwhelmingly successful, and transactions with observable contract logs represented a substantial share of included activity.

The clearest activity trough occurred on September 13. Elevated mean gas prices occurred on a different set of days, illustrating why transaction count, computational usage, and gas price should be examined together rather than treated as interchangeable measures of demand.

Material limitations

- Scope: Ethereum Mainnet only, over one short window. The study does not measure ecosystem-wide Layer 2 activity or longer-term adoption.
- Granularity: daily aggregates can conceal block-level and intraday variation. Export reconciliation cannot verify every underlying transaction or address.
- Identity and value: active addresses do not measure people, and transaction counts do not measure economic value or transaction usefulness.
- Fees and classification: gas_price summaries do not isolate fee components or paid costs. Logs and CREATE traces do not deliver an exhaustive EOA-versus-contract classification.
- Inference: endpoint changes, correlations, and z-score flags describe the sample. They do not establish causes or attribute observations to external events.

Targeted follow-up analysis

Prioritize September 13 for an hourly breakdown of transaction counts and address participation. Compare transaction-level gas distributions on low- and high-count days to test how computational cost per transaction changes. Inspect receipt and trace errors on higher-failure days. If resources permit, use historical account-state evidence for recipient classification and extend the window to evaluate weekday effects and anomaly stability.

These are proposed investigations, not explanations established by this report. The project demonstrates a reproducible analytical chain from SQL aggregation to validated metrics, charts, and appropriately bounded findings.

Reproducibility and data lineage

All primary tables belong to bigquery-public-data.goog_blockchain_ethereum_mainnet_us. Paths below are relative to the project root. The CSV exports are the analytical inputs; notebook filenames identify the corresponding analysis.

Task / source tables	SQL file	Notebook file
1 / transactions	sql/01_transaction_activity.sql	notebooks/01_transaction_activity.ipynb
2 / transactions	sql/02_active_addresses.sql	notebooks/02_active_addresses.ipynb
3 / reuse Tasks 1–2	sql/03_transaction_intensity.sql	notebooks/03_transaction_intensity.ipynb
4 / blocks	sql/04_network_utilization.sql	notebooks/04_network_utilization.ipynb
5 / transactions, blocks	sql/05_gas_activity.sql	notebooks/05_gas_activity.ipynb
6 / receipts	sql/06_transaction_success.sql	notebooks/06_transaction_success.ipynb
7 / transactions, logs, traces	sql/07_contract_vs_eoa.sql	notebooks/07_contract_vs_eoa.ipynb
8 / reuse Tasks 1–5	sql/08_network_activity_anomalies.sql	notebooks/08_network_activity_anomalies.ipynb
9 / reuse Tasks 1–5	sql/09_metric_relationships.sql	notebooks/09_metric_relationships.ipynb

Analytical input files

- data/processed/daily_transactions.csv — Task 1 and shared transaction counts.
- data/processed/daily_active_addresses.csv — daily sender-or-recipient union counts.
- data/processed/daily_network_utilization.csv — block-level daily aggregates.
- data/processed/daily_gas_activity.csv — price summaries and gas consumption.
- data/processed/daily_transaction_success.csv — receipt counts and outcome rates.
- data/processed/daily_contract_activity.csv — log and CREATE-trace measures.

Task 3 computes transaction intensity in Pandas by joining the first two exports on date. Tasks 8 and 9 also derive their results from existing exports. Their SQL files are methodological references, not additional extraction queries. A separate daily_transaction_intensity.csv is not required to reproduce these results.

sql/00_data_validation.sql records the window and block selection. For reproduction, preserve its fixed timestamps, use the source table fields shown in the SQL, export the daily results, and apply the explicit August 25–September 22 complete-day filter. Re-querying requires BigQuery access; estimate bytes processed before execution.

Figures 1–8 use the same daily exports and Pandas calculations as the report tables. The report package also includes a standalone active-address chart, report_02_addresses.png, for portfolio reuse.

Metric definitions

Metric / field	Definition and source	Interpretation boundary
date	UTC DATE(block_timestamp); all daily queries.	Timestamp filtering happens before daily grouping.
transaction_count	COUNT(*) from transactions, by date.	Included transaction count; not token transfers or users.
active_addresses	COUNT(DISTINCT address) over non-null from_address and to_address.	Daily union of senders and recipients; includes contracts.
transactions_per_active_address	transaction_count / active_addresses; Pandas.	Aggregate daily ratio, not address-level send frequency.
avg_transactions_per_block	AVG(transaction_count) from blocks.	Daily mean over included blocks.
total_gas_used	SUM(gas_used) from blocks.	Gas units consumed; not gas price or currency cost.
avg_gas_limit	AVG(gas_limit) from blocks.	Mean block gas limit in gas units.
avg_gas_utilization_pct	AVG(gas_used / gas_limit × 100), blocks.	Mean block-level ratio, not ratio of daily sums.
avg_gas_price_gwei	AVG(gas_price) / 1e9, transactions.	Field-based transaction mean; fee components not separated.
median_gas_price_gwei	APPROX_QUANTILES(gas_price,100)[50] / 1e9.	Approximate daily median, not exact.
success_rate_pct / failure_rate_pct	COUNTIF(status=1 or 0) / receipt count × 100.	Receipt execution outcome; not inclusion probability.
contract_interaction_transactions	Distinct transaction_hash with removed=FALSE in logs.	Observable log proxy, not all contract interactions.
contract_creation_transactions	Distinct transaction_hash with trace_type=create in traces.	Associated CREATE traces; not successful deployment count.
transactions_without_observed_logs	Transaction total minus transactions with non-removed logs.	Cannot be relabeled EOA-to-EOA activity.
z-score / coefficient of variation	(x - mean) / sample SD; CV = sample SD / mean × 100.	Within-window dispersion; no causal or significance claim.
Pearson correlation	Pandas correlation of complete-day metrics.	Linear association; shared inputs can create dependence.

Period shares use the relevant summed count divided by summed transaction count. Means of daily metrics weight each complete day equally. Values are rounded for display; all calculations use the unrounded exported inputs.